LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 3 Test: Metals and Non-metals

Chapter / Topic: Metals and Non-metals

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Which metal is a liquid at room temperature? **[1]**

(a) Sodium (b) Gallium
(c) Bromine (d) Mercury

Q2. Which of the following can be beaten into thin sheets? **[1]**

(a) Phosphorus (b) Zinc
(c) Sulphur (d) Oxygen

Q3. Which metal is stored under kerosene? **[1]**

(a) Iron (b) Copper
(c) Sodium (d) Aluminium

Q4. A metal oxide that reacts with both acids and bases to form salt and water is called: **[1]**

(a) a basic oxide (b) an amphoteric oxide
(c) an acidic oxide (d) a neutral oxide

Q5. Ionic compounds conduct electricity: **[1]**

(a) in the molten state or in aqueous solution
(b) in the solid state
(c) never
(d) only in the gaseous state

Q6. **Assertion (A):** Aluminium is more reactive than iron, yet aluminium articles do not corrode easily.
Reason (R): Aluminium forms a thin, protective layer of oxide on its surface. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Zinc gives hydrogen gas with dilute hydrochloric acid, but copper does not. Why? Write the balanced equation for the reaction of zinc. **[2]**

OR

Why is hydrogen gas usually not given off when a metal reacts with nitric acid? Name two metals that do give hydrogen with very dilute nitric acid.

Q8. What are amphoteric oxides? Give two examples. **[2]**

SECTION C

6 marks

Q9. Answer the following about the metals aluminium, copper, iron and zinc: **[3]**

(i) Arrange them in decreasing order of reactivity. **[1]**

(ii) Will a reaction take place when copper is added to iron(II) sulphate solution? Give a reason. **[1]**

(iii) Write the balanced equation for the reaction of zinc with copper sulphate solution. **[1]**

Q10. Mercury is obtained from its ore, cinnabar (HgS). Explain the steps with balanced equations. **[3]**

OR

What is a thermite reaction? Write the balanced equation and give one use.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Differentiate between roasting and calcination. Give one balanced equation for each. **[3]**

(ii) Give two methods of preventing the rusting of iron, and explain galvanisation. **[2]**

OR

(a) Why do ionic compounds have high melting points? [2]
(b) Ionic compounds are generally soluble in water but not in kerosene. Why? [1]
(c) Why do ionic compounds conduct electricity in the molten state but not in the solid state? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Pure gold, known as 24 carat gold, is very soft and is not suitable for making jewellery. It is alloyed with silver or copper to make it hard. In India, jewellery is usually made of 22 carat gold. An alloy is a homogeneous mixture of two or more metals, or of a metal and a non-metal. Brass is an alloy of copper and zinc, bronze of copper and tin, and solder of lead and tin. The melting point and electrical conductivity of an alloy are lower than those of the pure metals. **[4]**

(i) What does 22 carat gold mean? **[1]**

(ii) Name the metals in bronze. **[1]**

(iii) Why is solder used for welding electrical wires together? Give one more reason for making alloys. **[2]**

~ All the best! ~